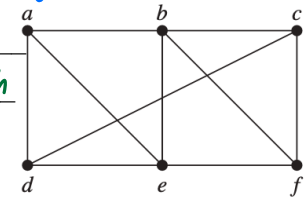
MAT2540, Classwork17, Spring2026

10.4 Connectivity

1. Introduction of Path:

   A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of graph. As the path travels along its edges, it visits the vertices along this path, that is, the endpoints of these edges.

2. Example of paths.



   (1) a, b, c, d, e, f is a path of length 5 since $\{a,b\}$, $\{b,c\}$, $\{c,d\}$, $\{d,e\}$, $\{e,f\}$ are edges in this graph

   (2) d, e, c, a is not a path since $\{e,c\}$ is not an edge in this graph

   (3) b, c, f, e, b is a path of length 4 since $\{b,c\}$, $\{c,f\}$, $\{f,e\}$, $\{e,b\}$ are edges

   (4) a, b, e, d, a, b is a path of length 5 and it goes through $\{a,b\}$ twice.

3. *Definition:* The **Path** of an **Undirected Graph**. Let n be a nonnegative integer and G an undirected graph.

   A *path of length* n from u to v in G is a sequence of n edges $e_1, e_2, \dots, e_n$ for which there exists a sequence of vertices u $x_0, x_1, x_2, \dots, x_n$ v such that $e_i = \{x_{i-1}, x_i\}$.

   When the graph is *simple*, we denote this path by its vertices $x_0, x_1, x_2, \dots, x_n$ because listing these vertices uniquely determines the path

   The path is *a circuit* if begins and ends at the vertex, that is, if $u = v$, and has length greater than zero.

   The path or circuit is said to pass through the vertices $x_1, x_2, \dots, x_n$ or traverse the edges $e_1, e_2, \dots, e_n$.

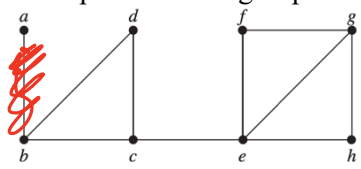
   A path or circuit is *simple* if it does not contain the same edge more than once

4. *Definition:* The **Subgraph** of A Graph

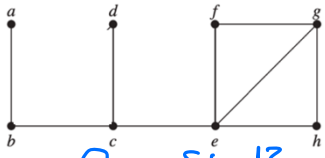
   A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$.

   A subgraph H of G is a proper subgraph of G if $H \neq G$.

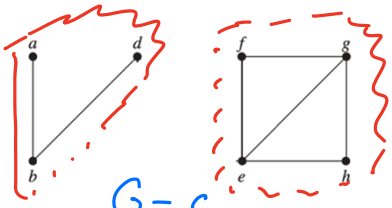
5. Example of the subgroup of a graph



G



$G - \{b,d\}$



$G - c$

6. *Definition:* The **Connected Graph**

   An undirected graph is called *connected* if there is a path between every pair of distinct vertices of the graph

   An undirected graph that is not connected is called disconnected. We say that we disconnect a graph when we remove vertices or edges, or both, to produce a disconnected subgraph

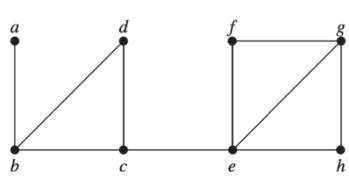
7. *Theorem:* There is a simple path between every pair of distinct vertices of a connected undirected graph.

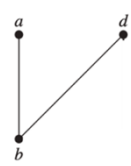
8. The **Connected Component of a Graph**

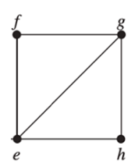
   If a graph G is not connected, then it has 2 or more connected components that are disjoint and have G as their union.

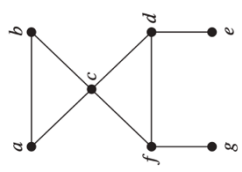
   Each connected component of G is not a proper subgraph of another connected component of G.

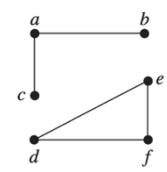
9. Examples. Check if each given graph is connected. If it is not connected, find its connected components.











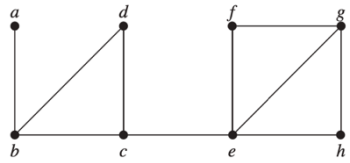
_____ _____ _____ _____

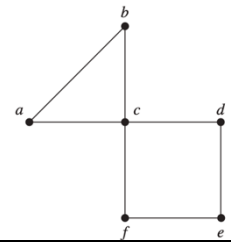
10. **Cut Vertex** and **Cut Edge**.

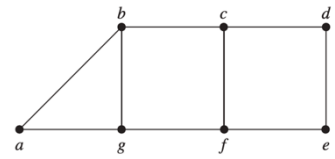
_____: A vertex is called *cut vertex* if _____ produces a subgraph with more _____. The removal of a cut vertex from a connected graph produces _____.

_____: An edge is called a *cut edge* or bridge if _____ produces _____ than in the original graph. If a graph has a cut edge, then we need only remove it to disconnect G.

11. Find the cut vertex and cut edge of each given graph.





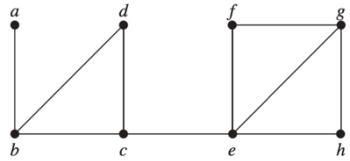


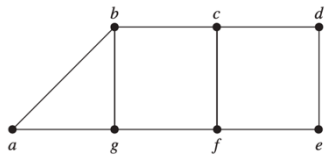
Cut Vertex			
Cut Edge			

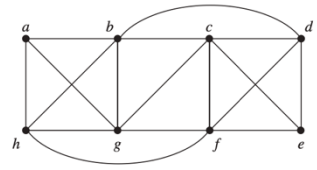
12. **Vertex Cut** and **Vertex Connectivity**.

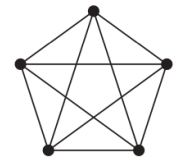
_____: A subset V' of the vertex set V of $G = (V, E)$ is a vertex cut, or separating set, if $G - V'$ is disconnected.

_____ of a noncomplete graph G, denoted by $\kappa(G)$, is _____ in a vertex cut.









Vertex cut _____ _____ _____ _____

Vertex connectivity _____ _____ _____ _____

To sum up, $\kappa(G)$ is minimum number of vertices that can be removed from G to either disconnect G or produce a graph with a single vertex. We have _____:

(1) _____ if G has n vertices, (2) _____ if and only if _____ or _____, and

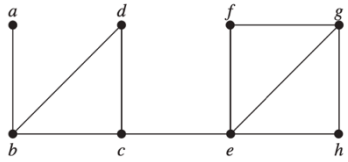
(3) _____ if and only if _____ with n vertices.

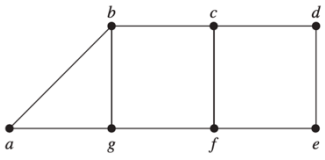
13. **Edge Cut** and **Edge Connectivity**.

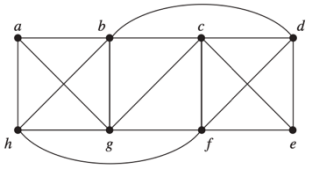
If G does not have a cut edge, we look for the smallest set of edges that can be removed to disconnect it.

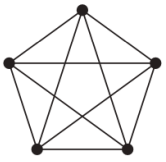
_____: A set of edges E' is called an edge cut of G if the subgraph $G - E'$ is disconnected.

_____ of a graph G, denoted by $\lambda(G)$, is the _____ in an edge cut of G.









Edge cut _____ _____ _____ _____

Edge connectivity _____ _____ _____ _____

To sum up, (1) _____ if G has n vertices, (2) _____ if and only if _____ or _____,

(3) _____ if and only if _____ with n vertices.

14. An Inequality For Vertex Connectivity and Edge Connectivity: For a graph $G = (V, E)$, we have

$$\kappa(G) \leq \lambda(G) \leq \min_{v \in V} \deg(v).$$